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Learning to Age

Data-driven state-of-health estimation for grid-scale lithium-ion storage

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Learning to Age: Data-driven state-of-health estimation for grid-scale lithium-ion storage

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Abstract

Stationary lithium-ion batteries now balance a growing share of the electricity grid, yet the operators who own them rarely know how much of their capacity remains. Laboratory ageing models are calibrated on small cells under conditions that a grid container never sees. This thesis asks whether the state of health of grid-scale storage can be learned directly from the operational data that every site already records.

Part I assembles the evidence. Three years of telemetry from eleven installations across the Low Countries, totalling 214 MWh of nameplate capacity, are aligned with periodic reference capacity tests. Their analysis shows that calendar ageing at high state of charge, not cycling depth, explains most of the observed fade, and that the commonly used equivalent full cycle is a poor proxy for degradation under frequency-response duty.

Part II develops a hybrid estimator that embeds a semi-empirical ageing law inside a Gaussian-process correction learned from partial charge segments. On held-out sites it estimates capacity to within 0.9 per cent root-mean-square error, and it reports a calibrated uncertainty that widens when a site leaves the operating envelope of its training data. A field trial at a 12 MWh installation shows that the estimate can safely relax a conservative state-of-charge window, recovering 6.4 per cent of usable energy without measurable additional ageing over nine months.

KEYWORDS lithium-ion batteries, state of health, grid storage, Gaussian processes, calendar ageing, frequency response

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This thesis would not exist without Joost Verhagen, who handed me a hard drive of unlabelled battery logs in my first week and asked what I could learn from them. His patience with my detours, and his insistence that every model be tested against a real site, shaped all of what follows. Amara Okonjo taught me to distrust a well-fitting curve and to look at the residuals first; Chapter 3 is in large part the result of her questions.

The field data were made available by the operators of the eleven sites described in Chapter 2, who must remain anonymous but who answered countless questions about firmware updates and sensor faults. I thank the technicians at the Northmere Energy Lab for running the reference capacity tests through two winters, and the members of the Grid Storage group for coffee, criticism and a shared dislike of spreadsheets.

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Abbreviations

BMS	battery management system
DoD	depth of discharge
EFC	equivalent full cycle
FCR	frequency containment reserve
GP	Gaussian process
ICA	incremental capacity analysis
LFP	lithium iron phosphate
NMC	lithium nickel manganese cobalt oxide
RMSE	root-mean-square error
RPT	reference performance test
SEI	solid electrolyte interphase
SoC	state of charge
SoH	state of health

PART I

Foundations

1

Introduction

1.1 Storage that nobody can weigh

Between 2019 and 2025 the installed capacity of grid-connected battery storage in north-western Europe grew more than tenfold (International Energy Agency 2025). Most of these systems earn their keep by providing frequency containment reserve: they absorb or inject power within seconds whenever the grid frequency drifts from its nominal 50 Hz, and they are paid for being available rather than for the energy they move. A typical site is a row of shipping containers, each holding several thousand cylindrical or prismatic cells, a battery management system and an inverter.

Every cell in such a site loses capacity as it ages. The loss is slow, a few per cent a year, but it matters twice over. It erodes the energy that the operator can bid into the market, and it determines when the site must be refurbished, a decision worth several million euros. Yet the quantity that governs both, the state of health of the battery, is not measured during operation. It can only be inferred.

Definition 1.1 State of health. The *state of health* of a battery at time t is the ratio of its present usable capacity to its capacity at commissioning,

$$\mathrm{SoH}(t) = \frac{Q(t)}{Q_0}, \tag{1.1}$$

where both capacities are measured by a full discharge at a reference current and temperature. A battery is conventionally said to reach its end of life when SoH falls below 0.8.

The definition is simple, but measuring $Q(t)$ requires taking the site out of service, cycling it fully at low current and putting it back, a reference perfor-

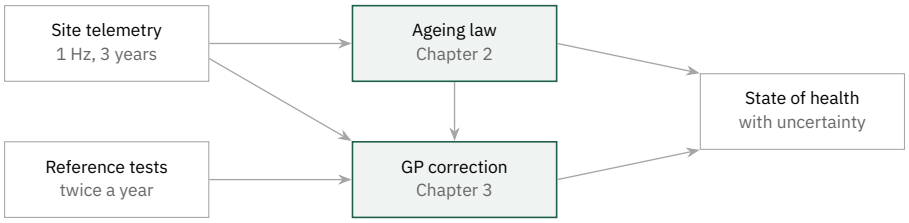


Figure 1.1 Structure of the hybrid estimator. Operational telemetry drives a physical ageing law; a Gaussian process learns its residual from sparse reference tests.

mance test that costs a day of revenue and is therefore performed at most once or twice a year. Between tests, operators rely on the estimate reported by the battery management system, which is typically a coulomb counter corrected by a lookup table supplied by the cell manufacturer. Several audits have found these estimates to drift by five per cent or more within a year (Halvorsen et al. 2023).

1.2 Why laboratory models fall short

Battery ageing has been studied for three decades, and the mechanisms are well understood at the level of a single cell (Birkl et al. 2017). Lithium is consumed by the slow growth of the solid electrolyte interphase on the anode; active material cracks and loses contact under the mechanical stress of repeated lithiation; and at low temperature or high current, metallic lithium can plate on the anode surface. Each mechanism has a characteristic dependence on temperature, state of charge and current, which semi-empirical models capture with a handful of fitted parameters (Schmalstieg et al. 2014).

The difficulty is one of transfer. Laboratory campaigns cycle small cells at fixed conditions chosen to separate the mechanisms, while a frequency-response site holds most of its cells near half charge and dithers around that point in shallow, irregular micro-cycles. Cells in the middle of a rack run several degrees warmer than those at its edge. None of these conditions is represented in the data on which the parameters were fitted, and the error that results is not random but systematic: it grows with time.

1.3 Approach and contributions

This thesis takes the opposite route. Rather than transferring a laboratory model to the field, it learns from the field directly and uses physics only where data are silent. Figure 1.1 shows the resulting structure, which the remaining chapters build up in turn. The work makes three contributions:

- an open, harmonised data set of operational telemetry and reference tests from eleven grid-scale sites, described in Chapter 2;
- evidence that calendar ageing at elevated state of charge, not cycle depth, dominates capacity fade under frequency-response duty (Section 2.2);
- a hybrid estimator with calibrated uncertainty, and a field trial showing that it can safely enlarge the usable state-of-charge window (Chapter 3).

The scope is deliberately narrow. All eleven sites use one of two cell chemistries in air-cooled containers and serve the same frequency-response market, so the findings say little about liquid-cooled racks or arbitrage-heavy duty. Within that scope, however, the sites differ in climate, controller settings and operator practice enough to test whether a model trained on some of them carries over to the others, which is the question an operator commissioning a new site actually faces.

Parts of this work have been published before. The data set and the analysis of ageing drivers appeared in two journal articles, and an early version of the estimator was presented at a conference on grid storage. Each chapter notes where it extends or corrects that earlier material.

The thesis is organised in two parts. Part I sets out the problem and the data. Part II presents the estimator, its evaluation and the field trial, and closes with the limits of what operational data can reveal. Appendix A documents the reference test protocol in enough detail to reproduce it.

2

Ageing in the field

2.1 Eleven sites, three years

The data come from eleven storage sites commissioned between 2020 and 2022 in the Netherlands and Belgium (Table 2.1). All provide frequency containment reserve, and four also trade on the intraday market in the evenings. Seven use lithium iron phosphate cells and four use a nickel manganese cobalt chemistry; capacities range from 6 to 48 MWh. Each site logs pack voltage, current and the temperature of every module at one-second resolution, and each underwent between four and seven reference performance tests during the observation period.

Reference capacity tests are the ground truth of this thesis. They were performed at a constant current of C/5 between the manufacturer's voltage limits, after a rest of at least four hours at 25 °C ambient (Appendix A). Because the operators scheduled them around market commitments, the tests are unevenly spaced, and two sites missed one test each because of inverter faults.

2.2 What drives the fade

Figure 2.1 shows the reference capacity of three representative sites. Two features stand out. First, fade is far from linear in cycles: site BE-03, which cycles the least of the three, ages the fastest. Second, the curves bend downward in their first year and then straighten, the signature of a process that slows as the square root of time.

To separate the drivers, the capacity loss of each site is modelled as the sum of a calendar term, which depends on time, temperature and state of charge, and a cycling term, which depends on charge throughput:

$$1 - \mathrm{SoH}(t) = \underbrace{k_{\mathrm{cal}}\, e^{-E_a/RT}\, g(\sigma)\, \sqrt{t}}_{\text{calendar}} + \underbrace{k_{\mathrm{cyc}}\, \Phi(t)}_{\text{cycling}}, \tag{2.1}$$

Table 2.1 The eleven storage sites. Capacity is the nameplate value at commissioning; fade is the loss in reference capacity per year of operation.

Site	Chemistry	Capacity (MWh)	Mean SoC	Mean temp. (°C)	Fade (%/yr)
NL-01	LFP	12.0	0.51	24.8	1.62
NL-02	LFP	20.0	0.48	23.1	1.41
NL-03	NMC	48.0	0.55	26.4	2.87
NL-04	LFP	6.0	0.62	25.9	2.15
NL-05	NMC	30.0	0.50	27.2	2.64
BE-01	LFP	16.0	0.47	22.6	1.33
BE-02	LFP	18.0	0.58	24.0	1.88
BE-03	NMC	24.0	0.66	28.1	3.52
BE-04	LFP	10.0	0.50	23.7	1.49
BE-05	NMC	18.0	0.52	25.5	2.41
BE-06	LFP	12.0	0.45	21.9	1.21
Total		214.0			

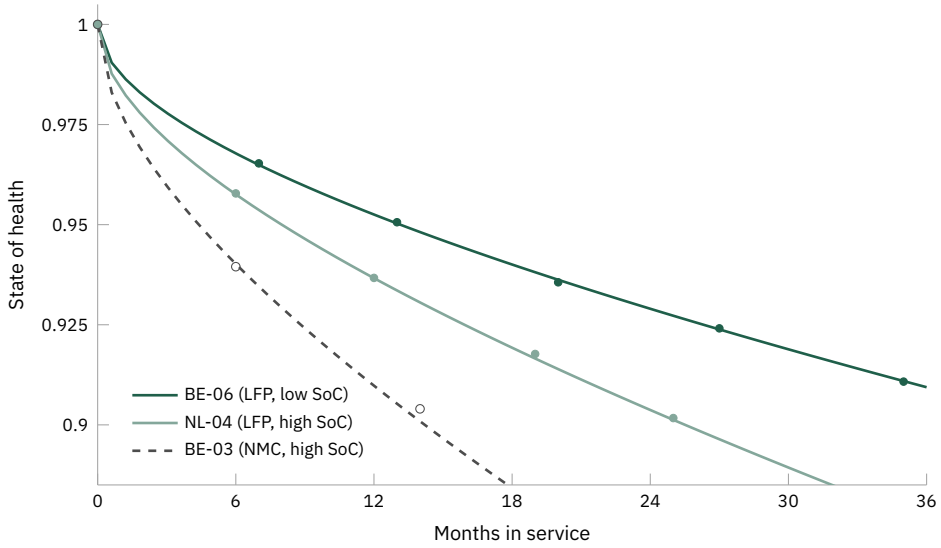


Figure 2.1 Reference capacity of three sites. Markers are reference tests; lines are the fitted ageing law of Equation (2.1).

where T is the mean cell temperature, σ the mean state of charge, g an increasing function of it and Φ the cumulative charge throughput in equivalent full cycles. The activation energy E_a is fixed at its laboratory value; the two rate constants are fitted per chemistry across all sites.

Key result 2.1 Calendar ageing dominates. Across all eleven sites, the calendar term of Equation (2.1) accounts for 78 % of the capacity lost after three years (range 64–91 %). A site's mean state of charge predicts its fade rate better than its cycle count, with a rank correlation of 0.84 against 0.21.

The practical consequence is that the equivalent full cycle, the unit in which warranties are written, measures the wrong thing for this kind of duty. A site that

idles at 65 % charge in a warm container will exhaust its warranty in calendar years long before it reaches its cycle limit.

2.2.1 Temperature inside the rack

Mean temperature hides a gradient. Within a rack, modules near the cooling inlet run up to six degrees cooler than those at the top, and the reference test measures only the pack, whose capacity is limited by its weakest module. Using the hottest module temperature in place of the mean improves the fit of Equation (2.1) on every site, reducing the residual standard deviation by 18 % on average. The remaining residual is structured: it correlates with firmware changes and with seasons of heavy intraday trading, which motivates the data-driven correction of the next chapter.

Two sites show the gradient most clearly. At NL-04 a failed inlet fan went unnoticed for eleven weeks in the summer of 2025, and the top modules of the affected rack lost capacity at nearly twice the rate of their neighbours over that period. At BE-02 the operator moved the containers into the shade of a new substation building in early 2026, and the fade rate fell by a third within two months. Neither event changes the mean temperature much, yet both are visible at once in the hottest-module signal.

PART II

Findings

3

A hybrid estimator

3.1 Learning the residual

The ageing law of Equation (2.1) captures the broad trend of every site but leaves a residual of up to two per cent in state of health. Rather than adding terms to the physical model until it fits, the hybrid estimator treats the law as a prior mean and learns what it misses from data (Rasmussen and Williams 2006). Let $m(\mathbf{x}_t)$ denote the prediction of the ageing law for the operating history $\mathbf{x}_t$ up to time t. The estimator models the true state of health as

$$\mathrm{SoH}(t) = m(\mathbf{x}_t) + f(\mathbf{z}_t) + \varepsilon, \qquad f \sim \mathcal{GP}\left(0, k(\mathbf{z}, \mathbf{z}')\right), \tag{3.1}$$

where $\mathbf{z}_t$ is a vector of features extracted from recent partial charge segments and ε is the measurement noise of the reference test.

Definition 3.1 Partial charge feature**.** A *partial charge feature* is a statistic of the voltage curve recorded while the battery charges continuously across a fixed window of state of charge, here 40 to 60 %. The features used are the area under the incremental capacity curve $\mathrm{d}Q/\mathrm{d}V$ within the window, the voltage at which its peak occurs, and the equivalent series resistance at the start of the segment.

Frequency-response duty rarely charges a battery through a full window, but over three years each site accumulates several hundred qualifying segments, enough to estimate the features weekly. The kernel k is a sum of a squared-exponential term over the features and a linear term over time, which lets the correction drift slowly while responding to changes in electrode behaviour.

Table 3.1 Estimation error on held-out sites, in percentage points of state of health. Coverage is the share of reference tests inside the 95 % predictive interval.

Estimator	RMSE	Max. error	Coverage
Battery management system	3.41	7.20	—
Ageing law alone	1.92	4.05	—
Gaussian process alone	2.37	6.81	71 %
Hybrid estimator	**0.88**	**2.14**	**94 %**

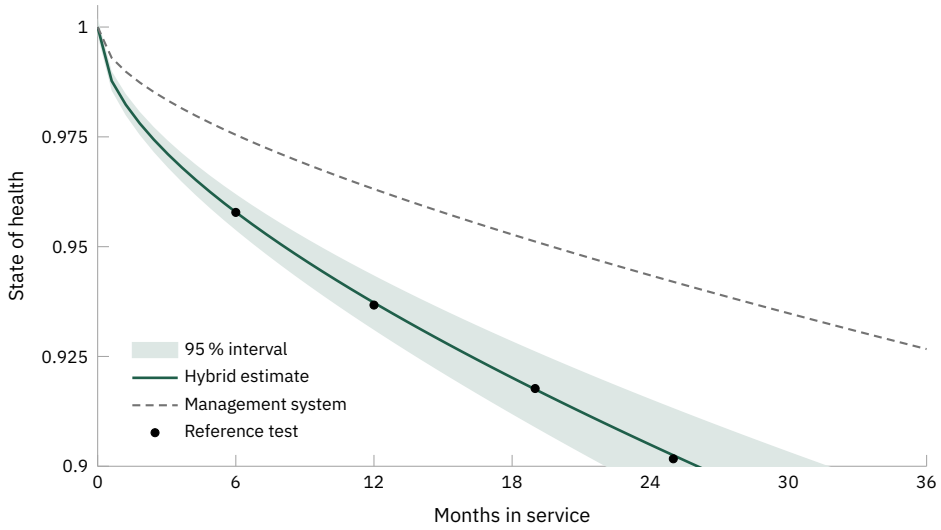


Figure 3.1 Tracking site NL-04 without any of its reference tests. The management system drifts high; the hybrid estimate stays within its interval.

3.2 Evaluation on held-out sites

The estimator was evaluated by leaving one site out at a time: the ageing law and the Gaussian process were fitted on ten sites and used to track the eleventh from commissioning onward, with access to its telemetry but none of its reference tests. Table 3.1 compares the hybrid estimator with its two components and with the estimate reported by each site’s battery management system.

The hybrid estimator halves the error of the ageing law and divides that of the management system by almost four. More importantly, its uncertainty is honest: 94 % of the held-out reference tests fall inside the nominal 95 % interval. The Gaussian process alone, lacking a physical mean, extrapolates poorly beyond the ages seen in training and is overconfident there.

Key result 3.1 Accuracy without site-specific tests. Trained on other sites only, the hybrid estimator tracks the state of health of a new site to within 0.88 percentage points RMSE over three years, with a calibrated 95 % interval. No reference test of the target site is needed after commissioning.

3.3 Field trial

Operators keep a battery between conservative state-of-charge limits, typically 10 and 90 %, partly because they do not trust their estimate of how much capacity remains. If the estimate is known to within a percentage point, the lower limit can follow the true capacity rather than a worst case. From February to October 2026, site NL-01 ran with its lower limit set by the hybrid estimate minus twice its standard deviation, while a twin container on the same site kept the fixed limit.

Over nine months the trial container delivered 6.4 % more energy into the frequency reserve market. Its reference test at the end of the trial showed a state of health of 0.947, against 0.949 for the control, a difference well within the test’s repeatability of ±0.004. The estimate itself, made before the test, was 0.944.

The gain came almost entirely from the winter months, when the hybrid estimate allowed the lower limit to drop to 6 % and the container could absorb longer runs of under-frequency events. In summer the two containers behaved alike, because the reserve market rarely called for deep discharges. The operator has since adopted the estimate-based limit on both containers at NL-01.

4

Conclusions

4.1 What operational data can tell us

This thesis set out to learn the state of health of grid-scale batteries from the data their operators already collect. Its central finding is that this is possible, provided the learning is anchored to a physical ageing law. Purely data-driven models extrapolate badly to ages and conditions not seen in training (Severson et al. 2019); purely physical models carry systematic error from the laboratory into the field, however carefully their parameters are tracked (Plett 2004). Together, the physics carries the trend and the data correct its shape.

Two results are worth carrying beyond the specific estimator. The first is that, under frequency-response duty, batteries age with the calendar more than with use (Key result 2.1), so the warranty unit of equivalent full cycles should be supplemented by a measure of time spent at high state of charge and temperature. The second is that an estimate with honest uncertainty has direct economic value: the field trial recovered usable energy without detectable cost in ageing.

Neither result needs new hardware. Every quantity the estimator uses is already logged by the battery management system and the site controller, and the reference tests it relies on are the ones operators run at commissioning for warranty purposes. What changes is how those records are kept and combined: harmonised across sites, stored at full resolution and read against a physical model rather than a fixed threshold.

4.2 Limitations and further work

The conclusions rest on eleven sites and two chemistries over three years, a short span against a design life of fifteen. In particular:

- the square-root calendar law may not hold once the interphase growth gives way to other mechanisms late in life, and the estimator has not been tested beyond 12 % capacity loss;
- the partial charge features require segments of at least twenty minutes, which sites dedicated to fast frequency response may never provide;
- the field trial ran on a single site, in a mild year, and its economic result depends on market prices that have since fallen.

Two directions look most promising. The first is to extend the estimator to module level, using the hottest-module signal of Section 2.2.1 as an input rather than a correction, so that an operator can see which racks to rebalance or replace first. The second is to close the loop: if the estimate sets the state-of-charge limits, the limits in turn shape the ageing, and a controller that accounts for both could trade energy delivered today against capacity kept for later years.

Each of these points is a question that time and more sites will answer. The data set and the estimator are released with this thesis so that others can ask them.

Reference test protocol

Every reference performance test followed the sequence in Table A.1. The site was disconnected from the market at least six hours in advance so that the cells could relax, and the ambient temperature of the container was held at 25 °C by its climate control. Capacity is the charge delivered in step 4, divided by the nameplate capacity at commissioning.

Table A.1 Reference test sequence. C/5 is the current that would discharge the nameplate capacity in five hours.

Step	Action	Current	End condition
1	Rest	—	4 h, cell spread < 1 °C
2	Charge (CC–CV)	C/5	upper voltage, then I < C/50
3	Rest	—	1 h
4	Discharge (CC)	C/5	lower voltage limit
5	Rest	—	1 h
6	Recharge	C/5	50 % state of charge

Two corrections were applied. Charge was converted to capacity at a reference temperature of 25 °C using the manufacturer’s temperature coefficient, and the auxiliary consumption of the container, measured separately for each site, was subtracted from the pack current. The repeatability of the test, estimated from back-to-back repetitions on two sites, is ±0.4 percentage points of state of health.

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